THE UNIVERSITY OF BRITISH COLUMBIA

Dear Dr. Cottingham,

Please consider our paper, ‘Current environments and evolutionary history shape forest temporal assembly’ for publication as an article in *Ecology*.

The timing of plant life history events—such as budburst and leafout—have shifted with climate change, with increasing concern for how these changes will impact species and how communities assemble in time. Species timings are, however, highly variable, especially in temperate systems where the effects of climate change are predicted to be greatest. To better understand how variability in species responses to key environmental cues impact the timing of leafout across North America, we performed a large-scale experiment with 47 forest species and over 1500 plants collected across four sites, allowing us to make inferences at the population, species and community-levels. In using a phylogenetic Bayesian model, we were also able to test the importance of evolutionary relationships in shaping species cue responses. We found systematic differences in species’ timings, with stable responses to temperature and daylength, which reflected species’ shared evolutionary histories, but not differences between populations. Our findings further our understanding of the factors that shape the timing of spring growth in forest communities and provide insight into how changes in temperature may impact temporal assembly with climate change.

Two reviewers highlighted the strengths of our analysis, having found the approach we took ‘elegant’ and that our work contributes ‘important insights’ to the field. Their comments, questions and suggested clarifications have greatly improved the manuscript. In response to the reviewer comments, we have provided greater detail in our methods and more explicitly defined several key terms throughout the manuscript. We also performed an additional analysis that further supports our key findings and their interpretation. We have revised the discussion to provide a more in-depth overview of the literature and a more detailed explanation of the implications and importance of our findings to forecasting and predicting changes in forest communities with climate change. Finally, we revised our figures to improve their clarity and interpretability.

We believe the new manuscript has been greatly improved, as discussed in our point-by-point responses. The manuscript is 4866 words with a 201 word abstract, and 5 figures. Our manuscript is not under consideration elsewhere. We hope you find it suitable for publication in *Ecology*, and look forward to hearing from you.

Sincerely,

Deirdre Loughnan

Deirdre Loughnan
Sentinels of Change Postdoctoral Fellow
Hakai Institute | UBC Department of Zoology